

Summary

AI/ML engineer who takes models from prototype to production. I built and run a live analytics platform on GCP (daily ETL, BigQuery, CI/CD) and LLM agent workflows in Python. I also work lower in the stack: a ~65% baking speedup on Metal and CUDA as incoming maintainer of KIRI Engine's Blender plugin, plus contributions to Thrift-attention (GPU MODE). First-author ML research, 9% over SoTA, presented at NASA SARI 2024. MSc AI at HKUST, graduating October 2026 and eligible to work in both the UK and Hong Kong. Now looking for an AI/ML engineer or applied scientist role, shipping ML products at scale.

Technical Skills

Languages – Python, C. C++, SQL, Rust, HTML, JavaScript
Technologies – Cuda, Metal, Pytorch, OpenCV, Sklearn, Git, Docker, Unix/Linux, Bash, Nix, Tailscale, ImageJ
Platforms – Google Cloud Platform, BigQuery, Google Earth Engine, Cloudflare, Hugging Face
Areas – Machine Learning, Computer Vision, GPU Programming, Computer Graphics, Numerical Optimisation, CI/CD, MLOps

Work Experience (Research and Internships):

Machine Learning Intern Mphasis AI and Applied Tech Lab at Ashoka

Aug 2023 - Jul 2025

Pytorch, Google Earth Engine, Python, Linux

- Used ML and time series analysis (LANDSAT), to enhance Indian forest fire classification accuracy by 9% over SoTA.
- Presented at NASA SARI Conference 2024 to delegates including Nobel Laureate David Skole and Dr. Krishna Vadrevu from NASA.

Machine Learning Intern Tiger Analytics Pvt. Limited

Oct 2024 - Dec 2024

Python, Sklearn, Numpy, Pandas

- Optimized Bayesian regression, additive, and multiplicative ML models, developing solutions for large-scale commercial data.
- Leveraged feature engineering, validation techniques, and numerical optimisation to improve forecasting accuracy in market mix modeling.

Teaching Assistant Ashoka University

Aug 2023 - Dec 2024

Machine Learning, Deep Learning, Computer Vision, OpenCV, Python, PyTorch

- Assisted in CS-3410: Introduction to Machine Learning and CS-4912: Computer Vision courses, engaging 120+ students.
- 50+ office hours, thereby reinforcing core ML, Deep Learning, and Computer Vision concepts.

Open Source Contributions and Personal Projects

Capstone Project: Efficient Rendering with Dynamic Gaussian Splatting (Research) – Python, Pytorch, Numpy

Jan 2026 - Present

- Working under Dr. SANDER Pedro (HKUST CS Department) and Dr. LI Ma (Netflix Eyeline Labs) on submission to TVCG Journal
- Rewrote rendering and baking operations to page VRAM and improve performance on consumer GPUs.

KIRI 3DGS Blender Addon Contributor (Open Source) – Python, Pytorch, Metal, Cuda

June 2026 - Present

- Merged changes into most popular blender addon to animate 3DGS, which has 1000+ active users.
- Upcoming *maintainer* of repository after contributing speedup ~65% for baking on Metal (Apple Silicon) and CUDA

GPU MODE Contributor (Open Source) – C++, Cuda, Nix

June 2026 - Present

- Thrift-attention – a project that offers ~30% inference speedup on NVIDIA Grace Blackwell with marginal performance degradation
- Wrote bindings in C++ and Nix code to register Thrift-attention as a Hugging Face Transformers backend

Advanced TFT Analytics Tool (<https://tft-app.hrsh.uk/>) – GCP, BigQuery, Cloudflare, Tailscale, CI/CD, ETL, Docker

Aug 2025 - Present

- Prototyped on Google Cloud Platform with autoscaling and BigQuery (as a learning exercise), then migrated to a self-hosted Linux server (Cloudflare Tunnel, Tailscale) after right-sizing infrastructure to the actual load.
- Built an automated ETL pipeline with CI/CD. Automated tests, parsing nested JSON (20000 matches/D) into a queryable schema

Multi-Agent Orchestration for Creative Poster Generation

Sep 2025 - Nov 2025

- Orchestrated LLM agents with LangGraph and OpenRouter to generate creative posters end to end.
- Added structured logging and modular tests for each stage of the orchestration pipeline.

Capstone Project: Predicting Forest Fire spread in Central India

Aug 2024 - Dec 2024

- Replicated DiffusionSat (Diffusion model specific to Satellite Data) to predict forest fire spread in central India.
- Applied Monte Carlo methods to predict forest fire spread.

3D Reconstruction of C. Elegans – Python, Pytorch, OpenCV, ImageJ

Jan 2024 - May 2024

- Built a pipeline for Image Processing, 3D reconstruction, and visualisation of C. Elegans (a microscopic roundworm) from SDCM images.
- Applied transfer learning, comparing a pretrained nnUNet segmentation against traditional CV methods.

Education

The Hong Kong University of Science and Technology Hong Kong (SAR), China

Sep 2025 – Oct 2026

Master of Science, Artificial Intelligence (CGA: 3.67/4.0)

Upon graduation will be eligible to work in Hong Kong and the UK

Ashoka University Sonapat, India

Aug 2021 - Jun 2025

Postgraduate Diploma in Advanced Studies and Research and Bachelor: Advanced Major in Computer Science

Coursework: Intro to Machine Learning, Computer Vision, Numerical Algorithms and Optimisation, Computer Graphics, Statistical Inference